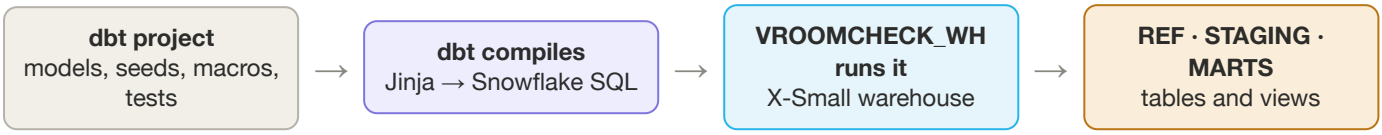
What dbt does in Snowflake

Vroomcheck · dbt Core 1.12 with the dbt-snowflake 1.12.4 adapter · database VROOMCHECK

> **In one sentence:** dbt runs on the laptop, but all the work happens inside Snowflake. It turns the project's Jinja SQL into plain Snowflake SQL, sends it over the connection, and Snowflake's warehouse runs it. The data never leaves Snowflake; this is ELT (load first, then transform inside the warehouse).



1 · How dbt connects

The `snowflake` target in `dbt/vroomcheck/profiles.yml` connects as follows:

Setting	Value
User	`VROOMCHECK_DBT`, a service user with key-pair login (no password)
Role	`TRANSFORMER`, which owns every object dbt creates
Warehouse	`VROOMCHECK_WH`, X-Small. It wakes when dbt sends a query and suspends 60 s after the last one
Database	`VROOMCHECK`
Threads	8, so up to 8 models build in parallel

This is dbt Core with the `dbt-snowflake` adapter, run from the laptop. It is not dbt Cloud, and not Snowflake's built-in dbt projects.

2 · What each command does in Snowflake

Command	SQL dbt sends to Snowflake	Result
`dbt seed`	`create transient table VROOMCHECK.REF.<seed>` plus `insert` batches from each CSV	5 REF tables. `severity_weight` is forced to `double`, otherwise Snowflake would read 0.3 as an integer and store 0
`dbt run` · staging	`create or replace view VROOMCHECK.STAGING.STG_<x> as select …`	4 views. Only the SQL is stored; Snowflake runs it when someone queries the view
`dbt run` · marts	`create or replace transient table VROOMCHECK.MARTS.<x> as (select …)`	14 tables, rebuilt from scratch every run ("create table as select")
`dbt test`	One `select` per test that returns the rows breaking the rule	0 rows = pass. 64 tests: 60 pass, 4 warn, 0 errors

`dbt build` runs all of this in dependency order, so seeds and staging come before the marts that read them. The last Snowflake build took **about 20 seconds in total**. The slowest models were `fct_complaints` (11 s), `fct_recalls` (11 s), `fct_tsbs` (9 s) and `bridge_tsb_vehicle` (9 s).

The actual compiled SQL for one mart, from `target/run/`:

```
create or replace transient table VROOMCHECK.marts.dim_vehicle as ( select … )
```

3 · What changes for Snowflake: the cross-database macros

The model SQL is identical on both engines. Where DuckDB and Snowflake need different functions, a macro picks the right one at compile time using `adapter.dispatch`:

Need	DuckDB version	Snowflake version dbt generates
Parse YYYYMMDD dates	`try_strptime(x, '%Y%m%d')`	`try_to_date(x, 'YYYYMMDD')`
Trim spaces	`trim(x)`	`trim(x, chr(32) || chr(160) || …)`, because Snowflake doesn't strip non-breaking spaces by default
Regex	`regexp_replace(…, 'g')`, `regexp_matches`	`regexp_replace`, `regexp_like`, `regexp_substr`
Every recall number in a narrative	`unnest(regexp_extract_all(…))`	`lateral flatten(regexp_substr_all(…))`
Distinct list of values	`string_agg(distinct …)`	`listagg(distinct …) within group (order by …)`
1970–2030 date spine	DuckDB's date-range function	`table(generator(rowcount => …))` with `dateadd`
Month name, ISO weekday	DuckDB functions	`to_char(d, 'MMMM')`, `dayofweekiso(d)`

That is why Snowflake ends up with the same tables as DuckDB: every one of the 23 REF, STAGING and MARTS objects matched row for row.

4 · Schema names

The `generate_schema_name` macro builds models straight into `STAGING`, `MARTS` and `REF`. Without it, dbt's default would have produced names like `ANALYTICS_STAGING`.

5 · What dbt does not do in Snowflake

- **Loading RAW:** dbt never writes to RAW. `load_raw_snowflake.py` loads it, and the 19-check gate checks it (results in `RAW._DQ_RESULTS`). dbt only reads it.
- **The ML layer:** dbt creates no ML tables in Snowflake; that layer is future scope.

6 · How to show it in the presentation

1. **Run it**, with the Snowflake values from `.env` loaded into the shell:

   ```
   cd dbt/vroomcheck && VC_DBT_TARGET=snowflake dbt build
   ```

2. **Query History:** in Snowsight, filter by user `VROOMCHECK_DBT`. Every `create or replace transient table …` and every test query dbt ran is listed, with its time and warehouse.
3. **Database explorer:** open `VROOMCHECK.MARTS` to show the tables dbt built.

One line for the presentation

> *“In Snowflake, dbt is the transformation engine. It compiles our models into Snowflake SQL, and Snowflake's warehouse runs them: seeds into REF, views in STAGING, transient tables in MARTS, then 64 tests. The full build takes about 20 seconds on an X-Small warehouse.”*